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==========

$a\triangleright: X \triangleleft F$

$a\triangleright: X \text{-}< F$

$a\triangleright: X \triangleleft F$

$aj\triangleright: X \to Fj \to X$ for each $j \in S$

$aj\triangleright = ai\triangleright \circ Fh$

$\triangleleft a: F \triangleright X$

$\triangleleft a: F >\text{-} X$

$\triangleleft aj: Fj \to X$ for each $j \in J$

$\triangleleft aj = Fh \circ \triangleleft ai$ for any $h: i \to j \in J$

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==== Cone ====

Cone

Let $F: J \to A$

Cone

$\triangleleft a: X \triangleleft F$

$\triangleleft aj: X \to Fj$ for each $j \in J$

$\triangleleft aj = \triangleleft ai \circ Fh$ for any $h: j \to i \in J$

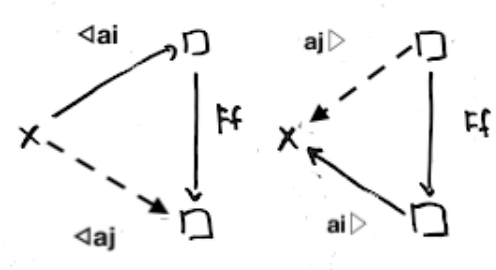
Co-cone

$a\triangleright: F \triangleright X$

$aj\triangleright: Fj \to X$ for each $j \in J$

$aj\triangleright = Fh \circ ai\triangleright$ for any $h: i \to j \in J$

Cone Co-cone



Composition

$\triangleleft a \circ f: X' \triangleleft F$ for $f: X' \to X$

$\sigma \circ \triangleleft a: X \triangleleft G$ for $\sigma: F \Rightarrow G$

$f \circ a\triangleright: F \triangleright X'$ for $f: X \to X'$

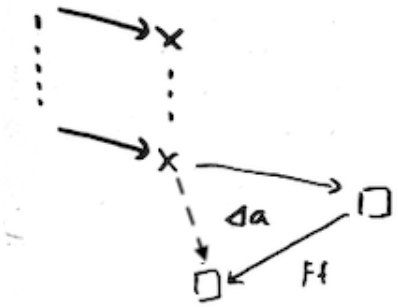
$a\triangleright \circ \sigma: G \triangleright X$ for $\sigma: G \Rightarrow F$

Base cone map

$\triangleleft a^* \in [_ \to X] \triangleleft F$ with base cone $\triangleleft F$

$\triangleleft a^*: \{_ \to X\} \to \{_ \triangleleft F\}$
$\triangleleft a^* (h) = \triangleleft a \circ h$

Base co-cone map
$^*a\triangleright \in F\triangleright[X \to _]$ with base co-cone $F\triangleright$
$^*a\triangleright: \{X \to _\} \to \{F\triangleright _\}$
$^*a\triangleright (h) = h \circ a\triangleright$



Proposition: natural transformation
Cone is equivalent to

Cone		**Natural transformation**
$\triangleleft a: X \triangleleft F$	<=>	$\alpha: F \Rightarrow \Delta X$
Base cone map		**Base map lattice**
$^*\triangleleft a \in [_ \to X] \triangleleft F$	<=>	$^*\alpha \in [_ \to X] \Rightarrow F$
		$^*\alpha j \in [_ \to X] \to Fj$ natural in j
		i.e. $^*\alpha j = {}^*Fh \circ {}^*\alpha i$
		$\alpha j = Fh \circ \alpha i$

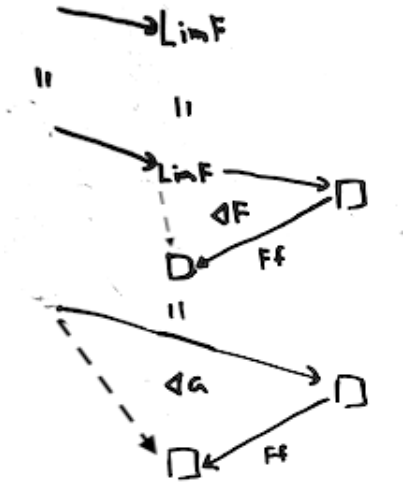
Limit (upper bound)
Colimit (lower bound)
If $^*\triangleleft F$ is an **isomorphic** co-base map, then the base point is a **limit**:

$^*\triangleleft F \in [_ \to \text{Lim } F] \triangleleft F$ limit

If $F\triangleright^*$ is an **isomorphic** base map, then the base point is a **colimit**:

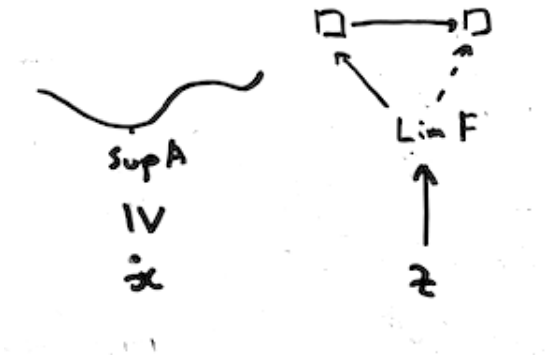
$F\triangleright^* \in F\triangleright[\text{Colim } F \to _]$ colimit

Limit



Note:

Limit ~ **sup**
i.e. least upper bound
Colim ~ **inf**
i.e. greatest lower bound

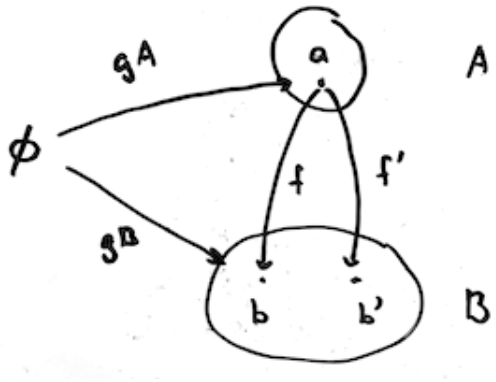


Trivial cone
Trivial limit/colimit
A limit is **trivial** if it is a **initial element**.
Analogously, a colimit is **trivial** if it is a **terminal element**.
In both case, there is only one **trivial cone**.

e.g.
Let
A = { a }
B = { b, b' }
J = {↓ | ↓} parallel arrows

with functor

F = { f | f' }
f, f': A → B
f(a) = b f'(a) = b'



If ◁g: X◁F is a cone, then

gA: X → A ◁gB: X → B
gB = f ∘ gA = f' ∘ gA

But if x ∈ X

gB(x) = f(gA(x)) = f(a) = b
gB(x) = f'(gA(x)) = f'(a) = b'

=> impossible, so

X = ∅
◁g: ∅◁F i.e. no map

Theorem: Limit functor
Suppose Lim F exists for all F: J→A
Then limit is a functor

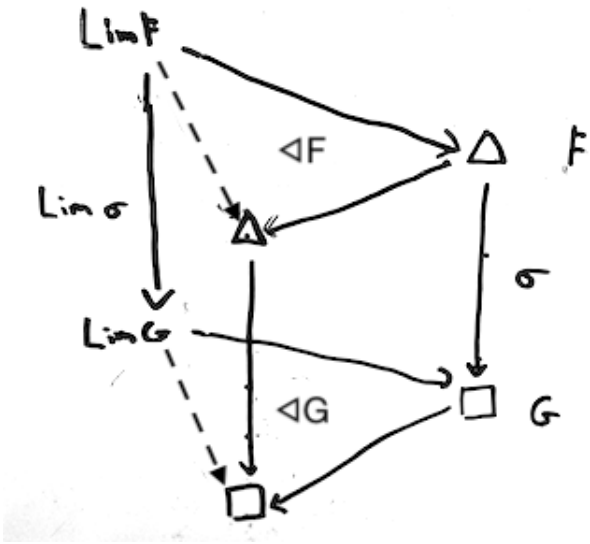
Lim: A^J→A
Lim F object map

Lim σ: Lim F → Lim G arrow map
where σ: F ⇒ G and unique Lim σ

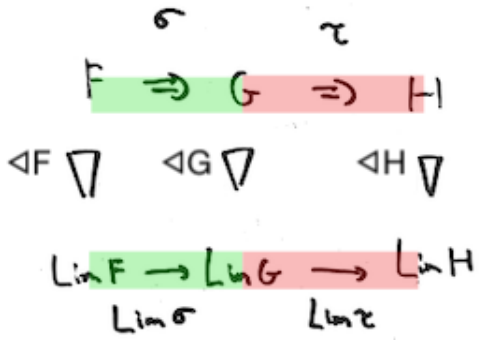
σ ∘ ◁**F** = ◁**G** ∘ **Lim** σ

Also, for X ∈ A

Lim X = Lim **Δ**X
Lim (◁a) = Lim (a: **Δ**X⇒F) : Lim X → Lim F
where ◁a: X◁F



Composition



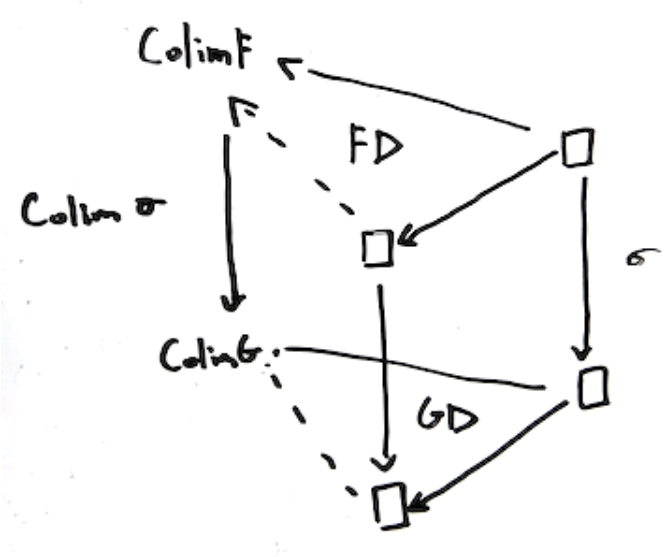
Theorem: Colimit functor
Suppose Colim F exists for all F: J → A
Then

Colim F: A^J → A

Colim σ: Colim F ⇒ Colim G
where σ: F ⇒ G
G▷ ∘ σ = Colim σ ∘ F▷

Let

Colim X = Colim ΔX
Colim (a▷: F▷X) = Colim (a: F ⇒ ΔX): Colim F → Clim X



==========

Proof

- Lim σ existence

Suppose σ: F ⇒ G. Then

$\sigma \circ \lhd F$: Lim F ⊲ G

=> $\sigma \circ \lhd F = \lhd G \circ \sigma'$ for a unique σ': Lim F→Lim G by Lim G

=> Lim σ = σ'

- composition: Lim τ σ = Lim τ ∘ Lim σ

Let σ: F ⇒ G, τ: G ⇒ H. Then

$\tau \circ \sigma \circ \lhd F = \tau \circ (\lhd G \circ$ Lim $\sigma)$ by Lim σ

$= (\lhd H \circ$ Lim $\tau) \circ$ Lim σ by Lim τ

$= \lhd H \circ$ (Lim $\tau \circ$ Lim σ)

=> Lim ($\tau \circ \sigma$) = Lim $\tau \circ$ Lim σ by uniqueness of Lim ($\tau \circ \sigma$)

QED

==========

Theorem: P-family limit

Limit of a family of functors is also a functor.

Let

F: J x P→ A family of functors

Fp = F(·,p): J → A

- **limit functor**: If each Fp has a limit, then the family of limits is a functor:

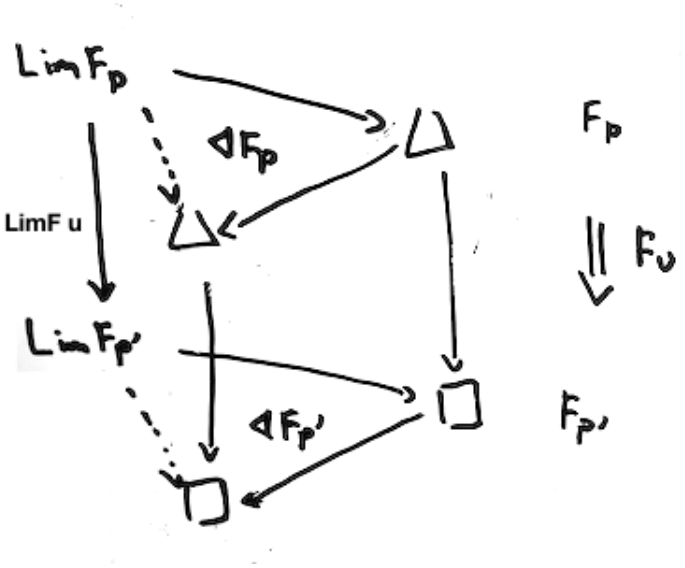
LimF: P → A

LimFp = Lim Fp object map for p ∈ P

LimFu: Lim Fp → Lim Fp' arrow map for u: p → p' ∈ P

such that Fu: Fp ⇒ Fp'

$\lhd Fp' \circ$ LimF u = Fu $\circ \lhd Fp$



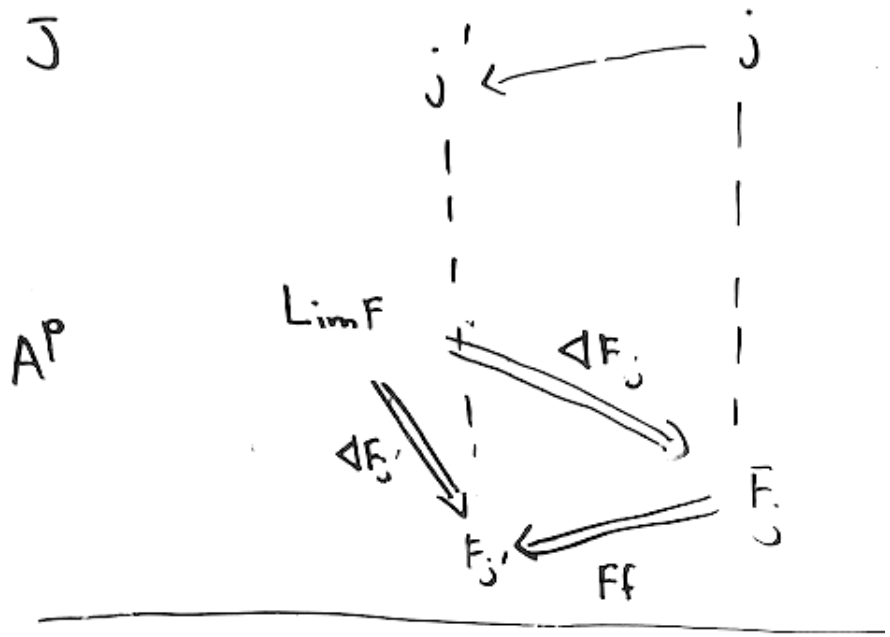
- **limit**: the limit functor itself is a limit of the P-family.
i.e. in the functor category A^P, LimP is a limit of F: J → A^P

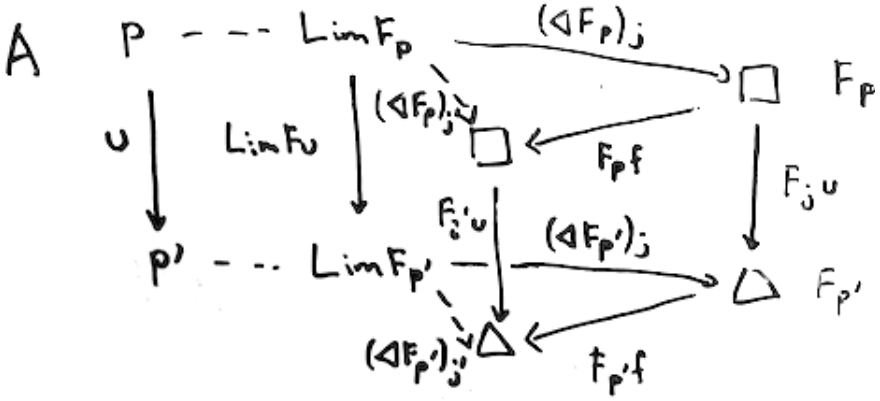
Limit
Lim F = LimF where F: J → A^P
LimF ∈ A^P

Limit cone
◁F: LimF ◁F
◁Fj: L ⇒ Fj

Corollary: continuity

Lim Fp = LimF p = (Lim F)p





==========

Proof

Idea: same as limit functor

- limit:

Pick any σ: T ⇒ F. Then for each p, there exists σ'p: Tp → Lim Fp such that

$\sigma p = \triangleleft Fp \circ \sigma' p$ by limit

claim: σ': T ⇒ L is well-defined

In particular, need to verify naturality.

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claim proof

Idea: use uniqueness of limits

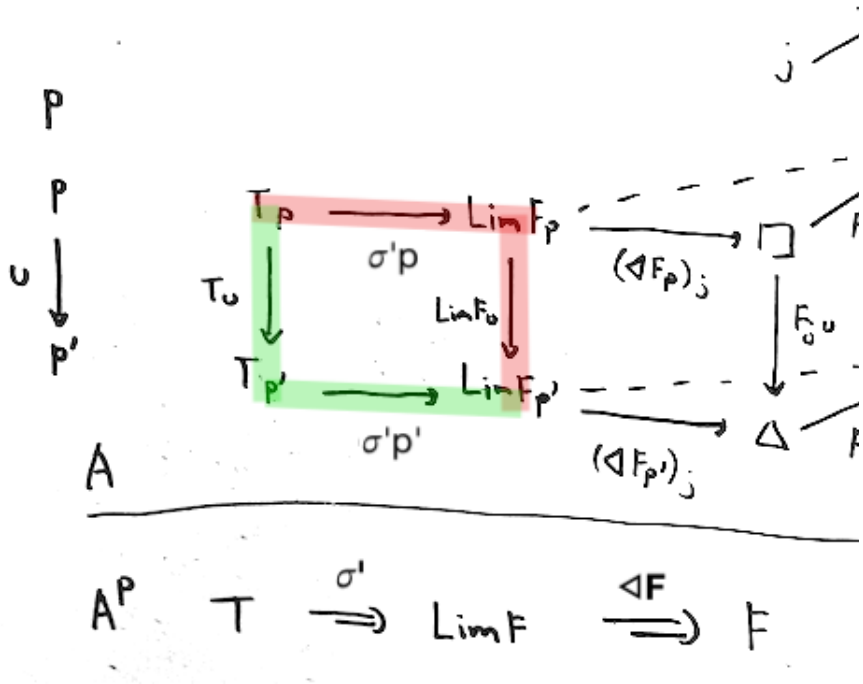
If u: p → p', then consider the cone Tp ◁ Fp'

$(\triangleleft Fp')j \circ (\sigma' p' \circ Tu) = ((\triangleleft Fp')j \circ \sigma' p') \circ Tu$

$= \sigma p' \circ TU$ by definition of σ

= Fju ◦ σp
= Fju ◦ ((◁Fp)j ◦ σ'p)
= (◁Fp')j ◦ (LimF u ◦ σ'p) by (◁Fp')j ◦ LimF u = Fju ◦ (◁Fp)j by definition of LimF

=> (σ'p' ◦ Tu), (LimF u ◦ σ'p): Tp → Lim Fp both generate Tp ◁ Fp'



=> σ'p' ◦ Tu = LimF u ◦ σ'p by uniqueness of limits
=> σ': T ⇒ L is valid

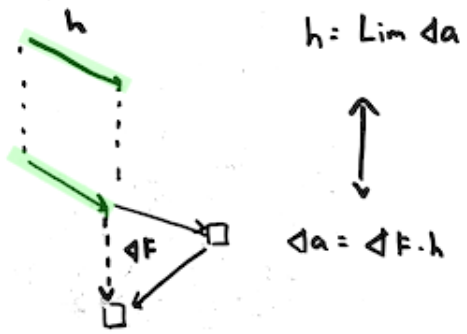
QED claim
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Hence σ = ◁F ◦ σ' by definition

QED
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Proposition: limit adjoint
If Lim F exists for every F:J→A (i.e. Lim is a functor), then *◁F ∈ [_→Lim F]◁F is an adjoint
That is, there is a natural 1-1 correspondence

Arrow		**Cone**
{ X → Lim F }	<=>	{ X ◁ F }
h: X →Lim F		◁a: X◁F
h = Lim ◁a		◁a = ◁F ◦ h



with

base map

◁F* ∈ [_→Lim F] ◁ F

◁F* (h) = ◁F ◦ h

co-base (inverse) map

Lim* ∈ Lim [X ◁ _] with limit functor Lim: A^J → A

Note: Lim X = X

Lim* (◁F ◦ h) = Lim (◁F ◦ h)

= h

Identity

◁F* ◦ Lim* (◁a) = ◁F * Lim (◁a)

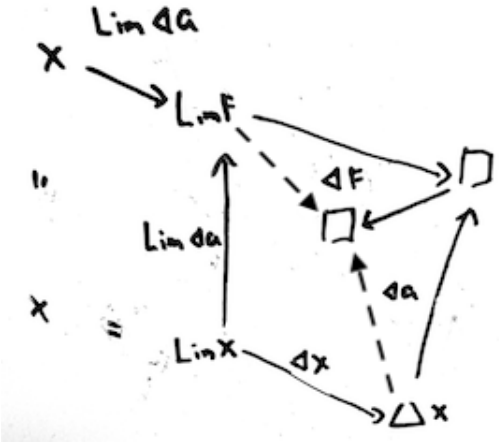
= ◁F ◦ Lim (◁a)

= ◁a

Lim* ◦ ◁F* (h) = Lim * ◁F (h)

= Lim (◁F ◦ h)

= h



Naturality

◁F'* ◦ *Lim σ = σ* ◦ ◁F* σ: F ⇒ F'

◁F' ◦ Lim σ = σ ◦ ◁F

with lattice

Base cone

◁: Lim ◁ I

◁F: Lim F ◁ F via whiskering ◁ * 1F = ◁F

Base co-cone

I: I ⇒ Lim **Δ** Identity

= I

Base identity

◁Lim ◦ Lim(I)

◁Lim ◦ Lim (F) = Lim (◁ (**Δ**Lim F))

=Lim F

Δ◁ ◦ I**Δ** = Identity

Δ◁ ◦ I**Δ** (X) = **Δ**(◁ **Δ**X)

=**Δ**X

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==== Preservation (aka continuity) ====

Limit preservation aka continuous

Limit creation

A functor F: A → B is **continuous** if it **preserves** base cones**.**

i.e. F is continuous if for any functor K: J → A

Lim K in A		**Lim FK** in B
◁K: Lim K◁K	=>	◁FK: Lim FK◁FK where
		Lim FK = F(Lim K)
		◁FK = F(◁K)

-

Dually, a functor F: A → B **creates limits** for A if it preserves cones in the opposite direction.

i.e. F creates limits for A if for any functor K: J → A

Lim FK in B		**Lim K** in A
◁FK Lim FK◁FK	=>	◁K: Lim K◁K where
		Lim FK = F(Lim K)
		◁FK = F(◁K)

e.g. the homset **{X→_}: A → Set** preserves limits (theorem below)

e.g. the forgetful **U: Grp → Set** creates limits (theorem below)

Proposition: composition

Let F: A → B, G: B → C

If F and G are continuous, then GF: A → C is continuous in C.

If F and G create limits, the GF: A → C creates limits for A.

Proposition: duality

If F: A → B creates limits and if a functor has at least one limit, then F also preserves limits for that functor.

i.e. If K: J → A and FK: J → B has a limit, then F preserves limits for K.

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proof:
Let ◁a: a◁K be a base cone of K created from FK i.e. F(◁a) is a base cone.
Then for any other base cone ◁a': a'◁K of K, there is an isomorphism f: a → a' such that

◁a' ◦ f = ◁a

=> F(◁a') ◦ Ff = F(◁a)
=> F(◁a') is isomorphic via Ff to a base cone F(◁a)
=> F(◁a') is a base cone.

QED
==========

Theorem: hom-functor continuity
Homset functors are coninuous
i.e. for a fixed X, let

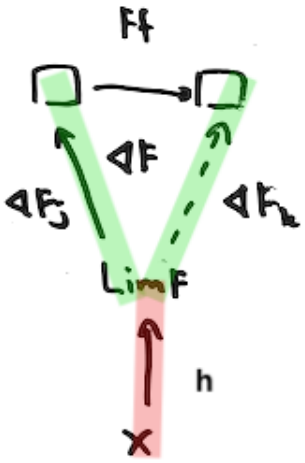
Hom-set functor
{ X → _ }: A → Set
f*: {X→Y} ⇒ {X→Y'} for any f:Y→Y'

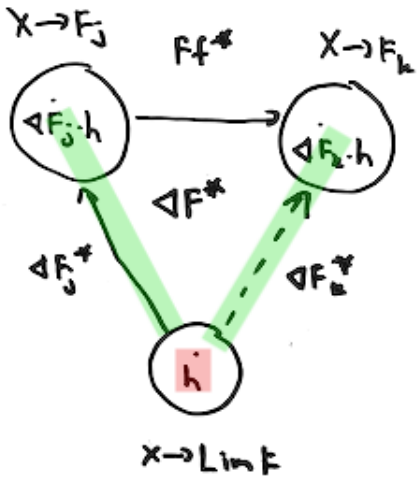
Then for any functor F: J → A with a limit, the induced functor F*: J → Set has limit

Limit
Lim F* = {X→Lim F}

Cone
◁F*: Lim F* ◁F*
◁F*j: {X→Lim F} → {X→Fj}
◁F*j = (◁Fj)*

A **=>** **Induced**





Note: the induced functor F* is

F*: J → Set
F*j = {X → Fj} object map for j ∈ J
Ff*: {X→ Fj} → {X→Fk} arrow map for f:j→k
Ff(h) = Ff ◦ h

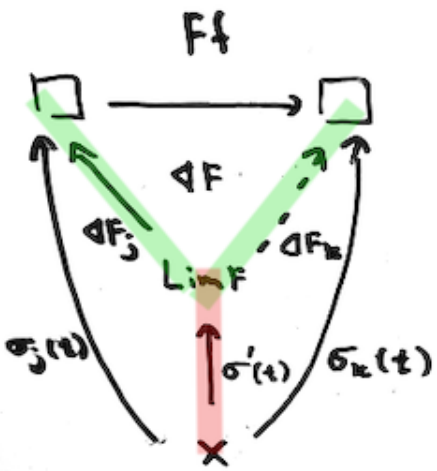
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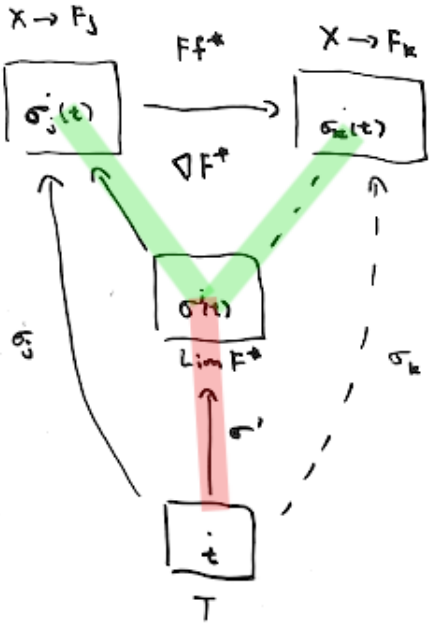
Proof
Let any σ: T◁F*.
Then for any t ∈ T

σj(t): X → Fj
σk(t) = Ff ◦ σj(t) for any f:j→k

=> σ(t) is a cone X◁F
=> there exists σ'(t): X→Lim F such that
σ(t) = ◁F ◦ σ'(t)
=> σ = ◁F* ◦ σ'

A => **Homset**





QED claim

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Proposition: adjoint continuity

Adjoints are continuous

i.e. Let

F: A→B G:B→A

K: J →B

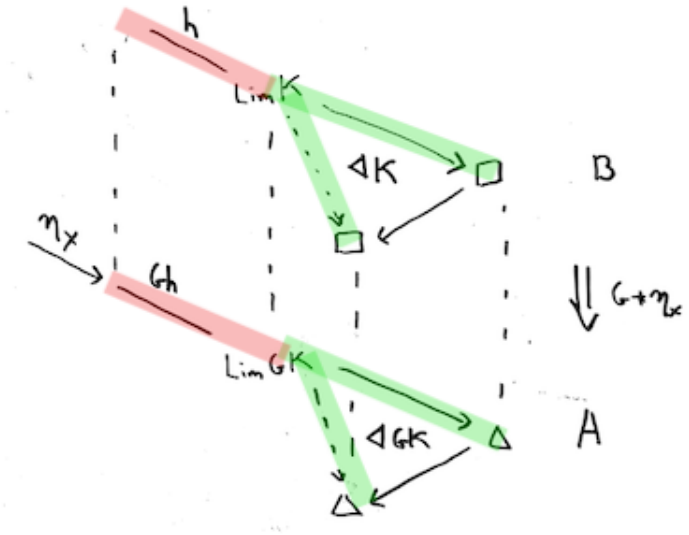
G * η ∈ ∗→G[F∗→_]

If G * η is an adjoint and Lim K exists, then Lim GK exists where

Limit	**Base cone**
Lim GK = G(Lim K)	◁(GK) = G(◁K)

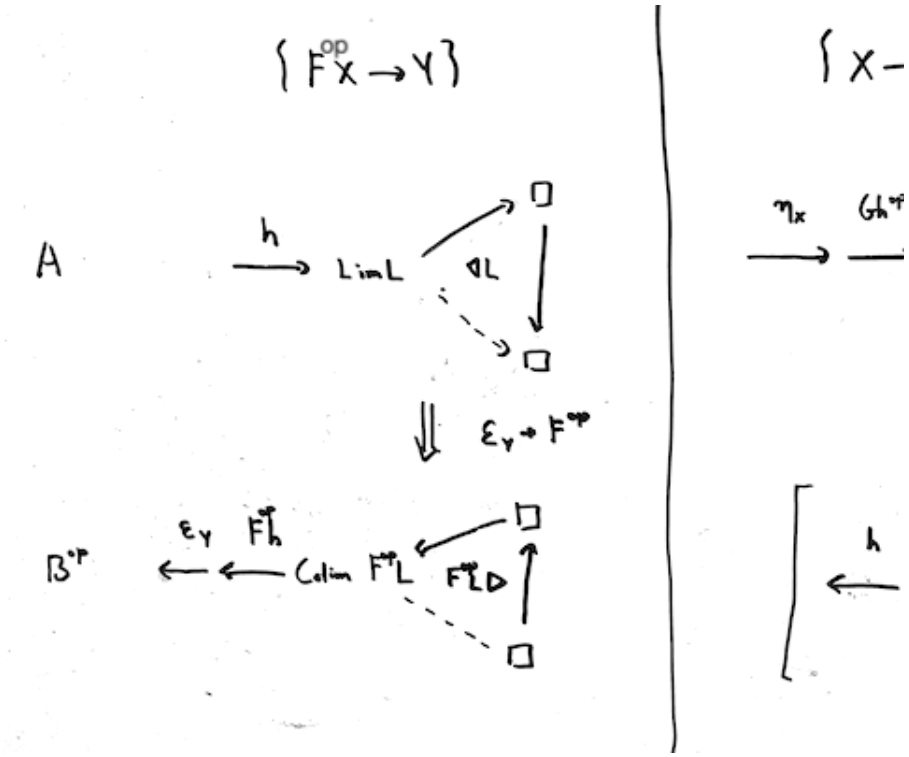
To be precise:

	◁(GK)* ∘ (G * η)	**=**	**(G * η) ∘ (◁K)***
base map	◁(GK) ◦ (Gh◦ ηX)	=	G[◁K ◦ h] ◦ ηX
Offset	Gh ◦ ηX: X → Lim GK	<=>	h: FX → Lim K



Corollary: opposites
Let functors
K: J → B
L: J → A

	B^op	**A**
Adjoint	{ F^opX → Y} F^op: A → B^op	{ X → GY } G: B^op → A
	limit of A as colimit in B^op	**colimit of B** as limit in A
Limit	Colim F^opL	Lim GK^op
	= F^op(Lim L)	= G(Lim K^op) = G((Colim K)^op)
Cone	(F^opL)▷	⊲(GK^op)
	= F^op(⊲L)	= G(⊲K^op) = G((K▷)^op)



	B	**A^op**
Adjoint	{ FX → Y } F: A^op → B	{ X → G^opY } G^op: B → A^op
	limit of A as colimit in B	**colimit of B** as limits in A^op
Limit	Colim FL^op = F(Colim L^op) = F((Lim L)^op)	Lim G^opK = G^op(Colim K)
cone	(FL^op)▷ = F(▷L^op) = F((◁L)^op)	◁(G^opK) = G^op(K▷)

==========

Proof Theorem

Idea: just by **composition** of isomorphisms ◁K* and G * η

i.e.

$G(\triangleleft K)^* \circ (G * \eta) = (G * \eta) \circ \triangleleft K^*$

Pick any $\triangleleft\sigma$: X◁ GK

Then ◁K* and G * η are both isomorphic, so

$\triangleleft\sigma = (G * \eta X) \circ \triangleleft K^* (h)$ for some unique h: X→Lim K

$\Rightarrow \sigma = (G * \eta X) \circ \triangleleft K^* (h)$

$= G(\triangleleft K)^* \circ (G * \eta X) (h)$ by composition

=> h' = G * ηX (h) is the unique arrow such that σ = G(◁K) (h')

Hence ◁(GK) = G(◁K)

QED
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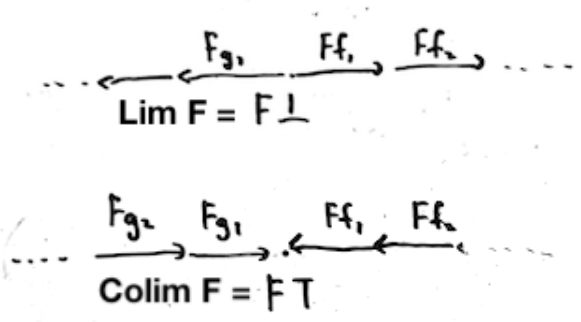
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==== Creation ====

Theorem: initial/terminal
Let F: J → A
If J has an initial object ⊤, then F has a limit

Lim F = F⊥
◁Fi: F⊤ → Fi

If J has an terminal object ⊥, then F has a colimit

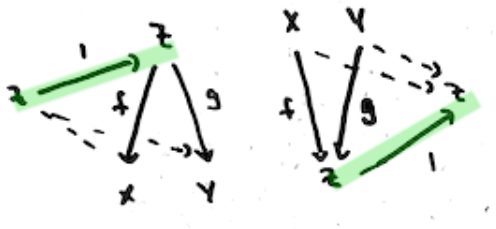
Colim F = F⊥
F▷i: Fi → F⊥



Corollary:

f: Z → X, g: Z → Y
Lim [f∧g] = Z
Colim [f∧g = X ⊔ Y

f: X → Z, g: Y → Z
Lim [f∨g] = X x Y
Colim [f∨g] = Z



==========
Proof:
- initial object ⊤
Suppose ◁a: X◁F. Let f = ◁a⊤: X → F⊤
Then for any j ∈ J, there is a unique

◁Fj: F⊤ → Fj

=> ◁aj = ◁Fj ∘ f

=> ◁a = ◁F ∘ f

QED
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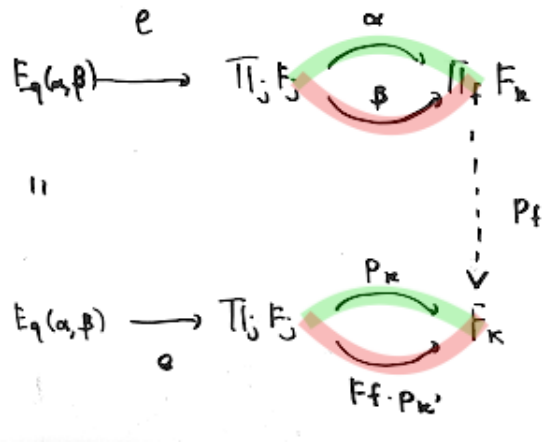
Theorem: limit
Let A be a category with all products and equalizers.
Then every functor has a limit.

To be precise, if F: J → A is a functor, consider the system of equations α, β: ∏j Fj → ∏f Fk for each f: k' → k

pf ∘ α = pk : ∏j Fj → Fk
pf ∘ β = Ff ∘ pk' : ∏j Fj → Fk

Equate them with the equalizer e: Eq(α, β) → ∏j Fj, so that

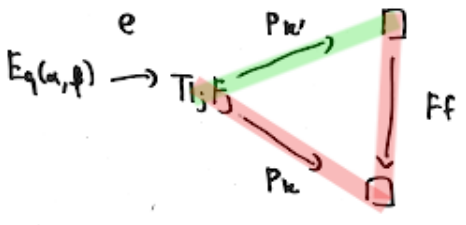
pk ∘ e = (pf ∘ α) ∘ e = (pf ∘ β) ∘ e = (Ff ∘ pk') ∘ e



Then the equalizer is the limit

Limit
Lim F = Eq (α, β)

Base
◁F: Eq (α, β) ◁ F
◁F = ∧j pj ∘ e where ◁Fj = pj ∘ e: Eq(α, β) → Fj for any j ∈ J



Theorem: colimit
If F: J → A a functor, consider the system of equations α, β: ⊔f Fk → ⊔j Fj for each f: k → k'

$\alpha \circ if = ik$: $Fk \to \sqcup j\ Fj$
$\beta \circ if = ik' \circ Ff$: $Fk \to \sqcup j\ Fj$

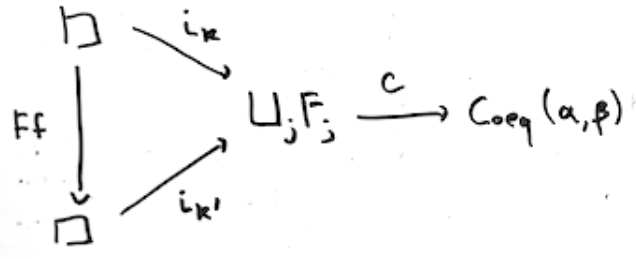
Equate them with the coequalizer c: $\sqcup j\ Fj \to Coeq(\alpha, \beta)$ so

$c \circ ik = c \circ (\alpha \circ if) = c \circ (\beta \circ if) = c \circ (ik' \circ Ff)$

Then the coequalizer is the colimit

Collimate
Colim F = Coeq (α, β)

Cobase
F$\triangleright$: F$\triangleright$ Coeq (α, β)
F$\triangleright$ = $c \circ \vee j\ ij$ where F$\triangleright$j = $c \circ ij$: $Fj \to Coeq(\alpha, \beta)$ for any $j \in J$



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Proof:

claim: $\sqcap j\ Fj \circ eq$ is indeed a cone
i.e. satisfies $Ff \circ (pj \circ eq) = (pj' \circ eq)$ for each $f: j' \to j$

proof claim:

$pj \circ eq$	=	$pf \circ \alpha \circ e$	by definition of α
	=	$pf \circ \beta \circ e$	by equalizer
	=	$(Ff \circ pj') \circ e$ by β	
	=	$Ff \circ (pj' \circ e)$	

QED claim

claim: $\wedge j\ pj \circ eq$ is a base

proof claim: Let any $\lhd a: Z \lhd F$
Let $a': Z \to \sqcap j\ Fj$ be the corresponding product form i.e.

$\lhd a = \wedge j\ pj \circ a'$

Then for any $f: j' \to j$

$pf \circ \alpha \circ a'$	=	$pj \circ a'$	
	=	$Ff \circ pj' \circ a'$	by $\lhd a$
	=	$pf \circ \beta \circ a'$	

=> a' equalizes (α, β)
=> a' = $eq \circ h$ unique $h: Z \to Lim\ F$
=> $\lhd a = \wedge j\ pj \circ (eq \circ h)$
$= (\wedge j\ pj \circ eq) \circ h$

=> ∧j pj ◦ eq is a base ⊲F

QED claim
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Corollary: complete sets

- **Set** is complete
For any functor F: J → Set, the limit is just the set of all combinations aka product that satisfy the triangle relation

Lim F = { x ∈ ∏j Fj: Ff(xj) = xj' for any f: j → j' }

⊲Fj (x) = xj jth projection

- Forgetful functor **U: Grp → Set** creates limits
Hence Grp is complete

Likewise for Vect, Mod, etc

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Proof

- **Set**: Set has products and equalizers
By the construction, Lim F is the equalizer of

α(x ∈ ∏j Fj) = ∏f Ff(xj)

β(x ∈ ∏j Fj) = ∏f xj' for f: j → j'

=> equalize

Lim F = Eq(α, β) = { x ∈ ∏f Fj: Ff(xj) = xj' for any f: j → j' }

- U: Grp → Set
Just need to verify that
a) Lim F can be made into a **group**
b) each ⊲Fj: Lim F → Fj can be made into a **homomorphism**

a) Let

product xy = ∏j (xj · yj)
inverse x^-1 = ∏j xj^-1

These are valid since

Ff(xy)j = Ff(xj)Ff(yj) Ff homomorphism
= xj'yj'
= (xy)j'

Ff(x^-1j) = Ff(xj)-1 Ff homomorphism
= xj^-1
= (x^-1)j

b) projection ⊲Fj is trivially homomorphic.

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==== Product/coproduct ====

Theorem: expressions
A product/coproduct is a way to represent multiple expressions as single.
If A has any products

fn: X → Yn for each n <=> f: X → ∏n Yn
fn = pn ∘ f

If A has any coproducts

gn: Xn → Y for each n <=> g: ⊔n Xn → Y
gn = g ∘ in

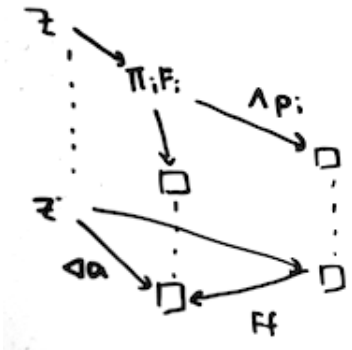
Theorem: product form
A cone can be expressed as a product
i.e. Let F: J → A.
If A has all products, then

◁a: Z◁F cone
a': Z → ∏n Fn product form

◁a = ∧n pn ∘ a' projection pm: ∏n Fn → Fm
◁an = pn ∘ a'

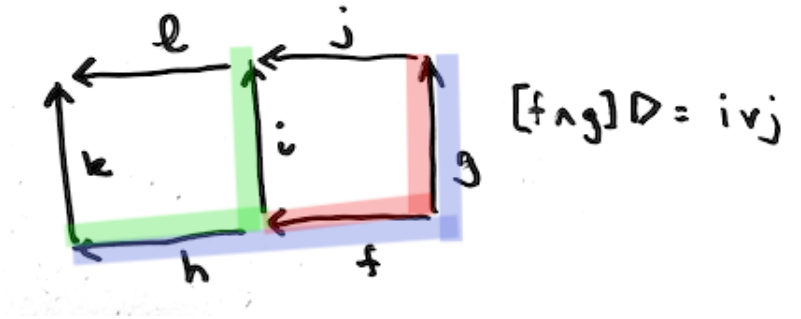
note: NOT every product is a cone since it needs to satisfy the triangle

pn ∘ a' = Ff ∘ (pn ∘ a')



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==== Pushout/pullback ====

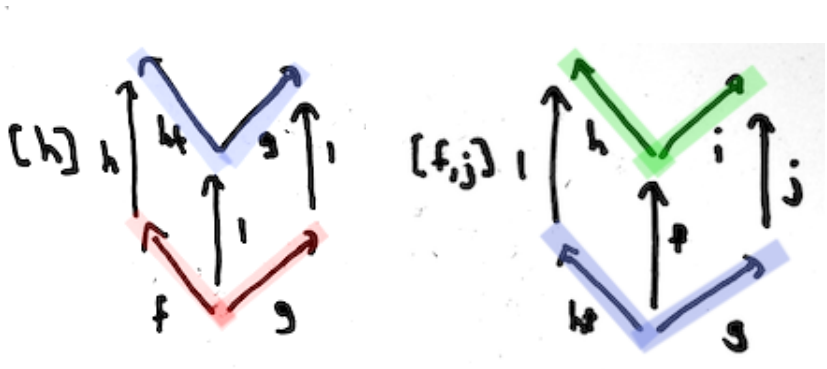
Proposition: pushout
Let squares
f ∧ g with colimit [f ∧ g]▷ = i ∨ j
h ∧ i



Related by natural transforms

[h]: [f ∧ g] ⇒ [hf ∧ g] where [h]_codom(f) = h

[f, j]: [hf ∧ g] ⇒ [h ∧ i] where [f, j]_dom(g) = f
[f, j]_codom(g) = j



a) If [h ∧ i] has a colimit with base [h ∧ i]▷ k ∨ l, then [hf ∧ g] also has a colimit.

Object
Colim [hf ∧ g] = Colim [h ∧ i]

[hf ∧ g]▷ = k ∨ lj

Transformation
Colim [h] = l

[hf ∧ g]▷ ∘ [h] = Colim [h] ∘ [f∧g]▷
k ∨ lj ∘ [h] = kh ∨ lj
= li ∨ lj
= l ∘ i ∨ j

b) If [hf ∧ g] has a colimit with base [hf ∧ g]▷ = k ∨ m, then [h ∧ i] has a colimit

Object
Colim [h ∧i] = Colim [hf ∧ g]

[h ∧ i]▷ = k ∨ l where l is the unique arrow kh ∨ m:

$[f \wedge g]\triangleright$ codom m

$kh \vee m = l \circ [f \wedge g]\triangleright$

i.e. **m = lj**

Transformation

Colim [f, j] = 1

$[h \wedge i]\triangleright \circ [f, j]$ = Colim [f, j] $\circ$ $[hf \wedge g]\triangleright$

= $[hf \wedge g]\triangleright$

$k \vee l \circ [f,j]$ = $k \vee lj$

= $k \vee m$

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proof:

a) Let any cone $a\triangleright = s \vee t: [hf \wedge g] \triangleright X$. Then

$sh \vee t: [f \wedge g]\triangleright X$

=> $sh \vee t = r \circ [f \wedge g]\triangleright$ for some unique r

$= ri \vee rj$

=> $sh = ri$ $t = rj$

Then also

$s \vee r: [h \wedge i]\triangleright X$

=> $s \vee r = r' \circ [h \wedge i]\triangleright X$ for some unique r'

$= r'k \vee r'l$

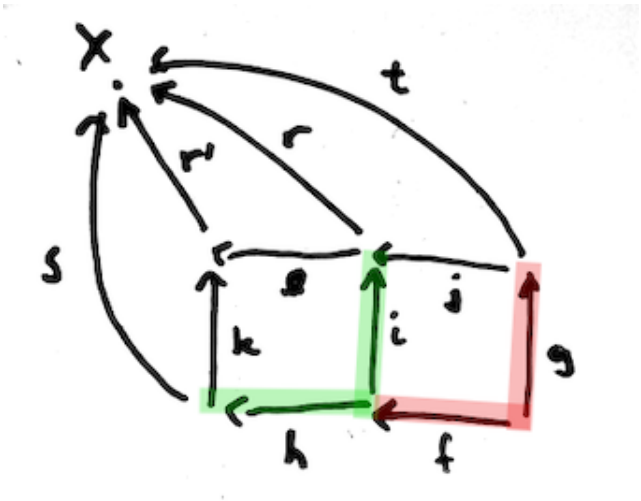
=> $s = r'k$

$t = rj = r'lj$

=> $a\triangleright = s \vee t$

$= r' \circ k \vee lj$

Hence $[hf \wedge g]\triangleright = k \vee lj$



QED

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==== OLD ====

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